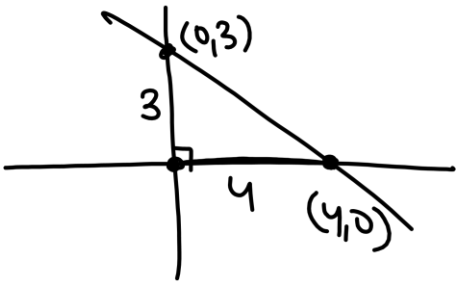
Intercept: (

x intercept, $y=0 \Rightarrow x=4$

y intercept, $x=0 \Rightarrow y=3$



1. Find the intercepts made by the line $3x + 4y - 12 = 0$ on the axes :
रेखा $3x + 4y - 12 = 0$ द्वारा अक्सों पर बनाये गये प्रतिछेदों को ज्ञात करो?

a) 2 and 3 b) 4 and 3
c) 3 and 5 d) NOT

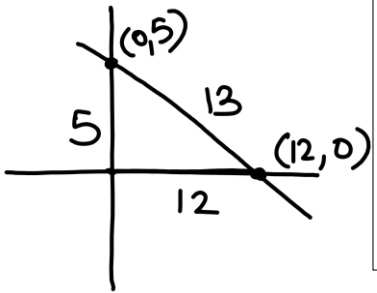
2. The shortest length of the intercept between the x-axis and y-axis of the equation $6x + 8y - 48 = 0$ is:
समीकरण $6x + 8y - 48 = 0$ के $x-$ अक्ष और $y-$ अक्ष के बीच अंतःखंड की सबसे छोटी लंबाई कितनी है?

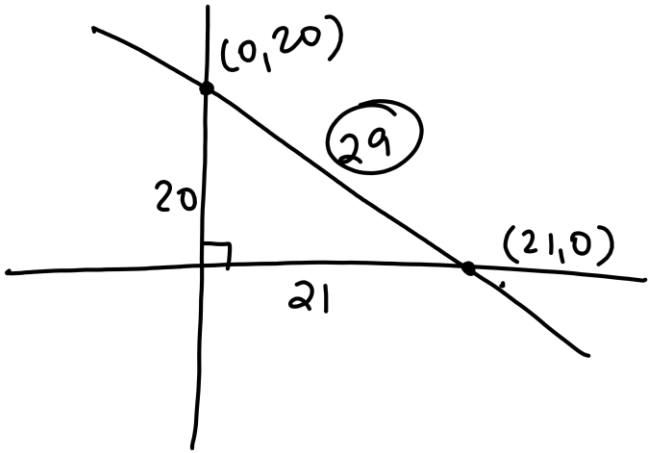
a) 10 units
b) 6 units
c) 14 units
d) 12 units

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3. **The length of the portion of the straight line $5x + 12y = 60$ intercepted between the axes is:**

a) 5 b) 13
c) 17 d) 12





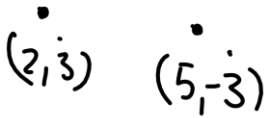
4. If the graph of the equation $20x + 21y = 420$ cuts the coordinate axes at P and Q, then what is the measure of the length (in units) of PQ?
यदि समीकरण $20x + 21y = 420$ का ग्राफ निर्देशांक अक्षों को P और Q पर काटता है, तो PQ की लंबाई की माप (इकाइयों में) क्या है?

a) 25 b) 29
c) 40 d) 35

Slope of a line
(

Slope/gradient of a line: ( )

- **Slope of a line represents its degree of inclination.**



$$\frac{6}{-3} = -2$$

iii) $ax + by + c = 0$

$$\text{Slope} = -\frac{a}{b} = \frac{-(\text{Coeff. of } x)}{\text{Coeff. of } y}$$

Eg: $4x + 7y - 18 = 0$

$$m = \frac{-4}{7}$$

- **Slope of $x - axis$ is 0.**
- **Slope of $y - axis$ is not defined.**
- **Slopes of two parallel lines are same $i.e. m_1 = m_2$**
- **Slopes of two perpendicular lines are negative reciprocals of each other $i.e. m_1 = -\frac{1}{m_2}$ or $m_1 \times m_2 = -1$**

5. **Find the slope of the line joining the points (7, 5) and (9, 7):**

a) 1 **b) 2**

c) $\frac{1}{2}$ **d) 3**

$$\frac{7-5}{9-7} = \frac{2}{2} = 1$$

$m = \tan\theta \rightarrow 45^\circ = \frac{\pi}{4}$

$\frac{2-1}{2-1} = \frac{1}{1} = 1$

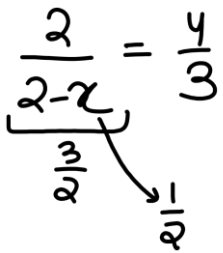
$\tan\theta = \frac{2}{3}$

$\theta = \tan^{-1}\frac{2}{3}$

6. **What is the slope of line joining the points $(1, 1)$ and $(2, 2)$?**
$(1,1)$ और $(2,2)$ के मिलाने से बनी रेखा का ढलान ज्ञात करो|

a) $\frac{\pi}{2}$ b) $\frac{\pi}{4}$

c) $\frac{\pi}{6}$ d) $\frac{\pi}{8}$



7. Slope of the line AB is $4/3$. Co-ordinates of points A and B are $(x, -5)$ and $(2, -3)$ respectively. What is the value of x?

रेखा AB की ढलान $4/3$ है। बिन्दुओं A और B के निर्देशांक क्रमशः $(x, -5)$ और $(2, -3)$ है । x का मान क्या है

a) $1/2$ b) $-1/4$

c) 4 d) -4

8. What is the slope of the line $5x + 3y = 12$
रेखा $5x + 3y = 12$ का ढाल क्या है?

a) $2/5$ b) $3/5$

c) $-3/5$ d) $-5/3$

$-\frac{5}{3}$

$+\frac{a}{3} = +\frac{2}{3}$

9. $ax + 3y = 6$ has slope $-2/3$ What is the value of a?

a) -2 b) 2

c) 3 d) -3

$$\frac{3-(-3)}{6-2} = \frac{6}{4} = \frac{3}{2}$$

10. A line passes through the points A $(2, -3)$ and B $(6, 3)$. Find the slope of the line which is parallel to AB ?
बिन्दुओं A $(2, -3)$ और B $(6, 3)$ से एक रेखा गुजर रही है। तो उस रेखा के समांतर रेखा की ढलान क्या है?

a) $-2/3$ b) $3/2$
c) $1/2$ d) $3/4$



11. What is the slope of the line perpendicular to the line passing through the points $(3, 5)$ and $(-4, 2)$?
उस रेखा का ढाल बताइए जो बिन्दुओं $(3, 5)$ और $(-4, 2)$ से गुजरती हुई रेखा के लम्बवत है?

a) $-3/5$ b) $-2/3$
c) $-7/3$ d) $7/3$

12. **What is the slope of line perpendicular to the line $2x - 5y = 12$**

a) $2/5$ b) $5/2$

c) $-2/5$ d) $-5/2$

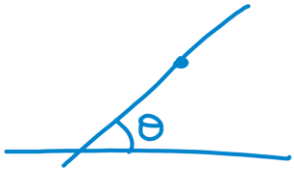
$$m_1 = \frac{-2}{-5} = \frac{2}{5}$$

$$m_2 = -\frac{5}{2}$$

Equation of a line
(

Different ways to form equation of a line:

- Slope point form
- Two-point form
- Intercept form
- Slope intercept form





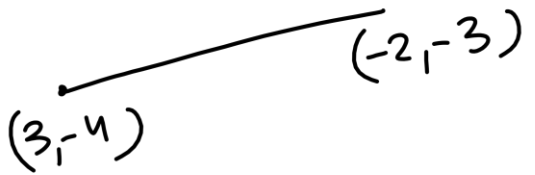
Slope point form:





Two-point form:

ii)



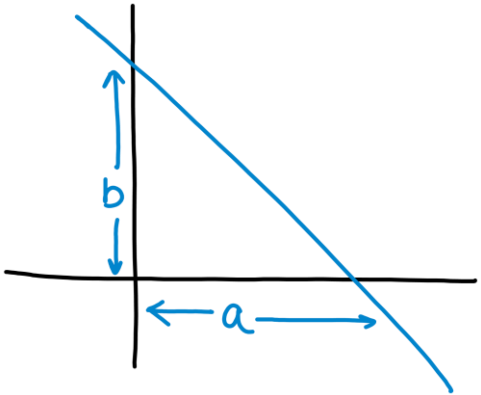
$m = \frac{-1}{5}$

$x + 5y = -17$

3-20

Intercept form:

$$\frac{x}{a} + \frac{y}{b} = 1$$

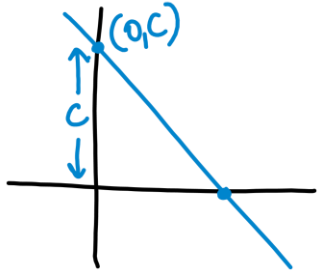


Slope intercept form:

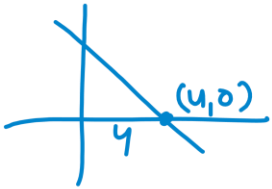
m

y intercept $= c$

$$y = mx + c$$



i) Ex $m = \frac{7}{3}$



$-7x+3y=-28$

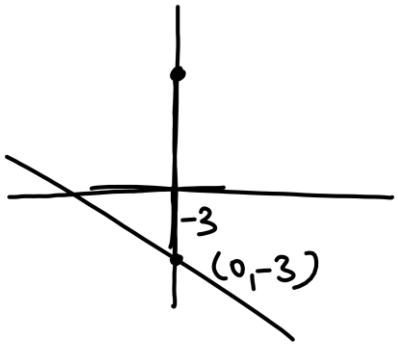
$-28+0$

$7x-3y=28$

ii) $m = \frac{5}{3}$

$y = mx + c$

$y = \frac{5}{3}x - 3$



$-5x + 3y = -9$

$0 - 9$

$$m = \frac{4}{-4} = \frac{-1}{1}$$

$$x + y = -3$$

$$-1 - 2 = -3$$

13. **Find the equation of the line through the points $(-1, -2)$ and $(-5, 2)$:**

a) $2x + y = 3$

b) $3x - 2y + 7 = 0$

c) $x + y + 3 = 0$

d) NOT

14. **Find the equation of the line joining the points of intersection of $2x + y = 4$ and $x - y + 1 = 0$ with point of intersection of $2x - y - 1 = 0$ and $x + y - 8 = 0$:**

a) $2x + 3y - 6 = 0$
b) $3x + 2y - 1 = 0$
c) $3x - 2y + 1 = 0$
d) None of these

$-5x+4y=-22$

$-10-12$

15. **Find the equation of the line passing through the point $(2,-3)$ and having its slope $5/4$:**

a) $4x-5y=23$
b) $10x-8y=5$
c) $5x-4y=22$
d) NOT

16. Find the equation of the line which passes through the point $(3, -4)$ and makes an angle of $60°$ with the positive direction of $x - axis$:

उस सरल रेखा का समीकरण ज्ञात कीजिये जो बिंदु $(3, -4)$ से गुजरती है और $x-$अक्ष की धनात्मक दिशा के साथ $60°$ का कोण बनाती है।

a) $3x - \sqrt{3}y = 4$

b) $\sqrt{3}x - y = 4 + 3\sqrt{3}$

c) $x\sqrt{3} + y = 3\sqrt{2} + 5$

d) NOT center

$m_1 = \frac{6-2}{5-1} = \frac{4}{4} = 1$

$m_2 = -\frac{1}{1}$

$x+y = -9$

$-4-5$

17. **Find the equation of the line passing through the point $(-4, -5)$ and perpendicular to the line joining the points $(1, 2)$ and $(5, 6)$:**

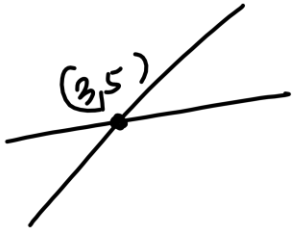
बिंदु $(-4, -5)$ से गुजरने वाली उस रेखा का समीकरण बताइए जो की बिन्दुओं $(1, 2)$ और $(5, 6)$ को मिलाने से बनने वाली रेखा के लम्बवत है?

a) $x + y + 17 = 0$

b) $3x + 2y + 11 = 0$

c) $x + y - 2 = 0$

d) NOT



$m_1 = \frac{6-4}{5-2} = \frac{2}{3}$

$m_2 = -\frac{3}{2}$

$3x + 2y = 19$

$9 + 10 = 19$

~~$5x = 15$~~ 3

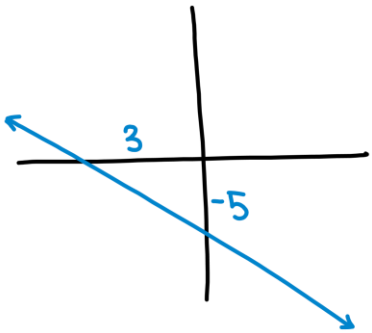
$y = 5$

18. **Find the equation of the straight line which passes through the point of intersection of the straight lines $x + y = 8$ and $3x - 2y + 1 = 0$ and is perpendicular to the straight line joining the points $(2, 4)$ and $(5, 6)$:**

a) $3x - 2y - 19 = 0$
b) $3x + 2y = 19$
c) $6x - 4y + 8 = 0$
d) NOT

$$\frac{x}{3} + \frac{y}{-5} = 1 \quad \Big\} \times 15$$

$$5x - 3y = 15$$



19. What is the equation of a line which has 3 as $x-$ intercept and -5 as $y-$intercept?
एक रेखा के समीकरण का पता लगायें, जिसकी $x-$ प्रतिच्छेद 3 और y प्रतिच्छेद 5 है?

a) $3x - 5y = 15$
b) $5x - 3y = 15$
c) $5x + 3y = 15$
d) $3x + 5y = 15$

20. **Find the equation of the line which cuts off intercepts 2 and 3 from the axes:**

a) $9x - 7y = 6$
b) $3x + 2y = 6$
c) $4x + 3y = 7$
d) NOT

$$\frac{x}{2} + \frac{y}{3} = 1 \quad \Big\} \times 6$$

$$3x + 2y = 6$$

